Zeyu Zhu

The Hong Kong University of Science and Technology, Clear Water Bay, Kowloon, Hong Kong

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I received my Ph.D. degree through the direct Ph.D. program at the Institute of Automation, Chinese Academy of Sciences, where I was affiliated with the Efficient Intelligent Computing and Learning Lab. My research focuses on **AI chip design**, **ML system design**, **heterogeneous computing**, and **neural network model compression** for LLMs, GNNs, and CNNs. To date, I have published nin first-author/co-first-author papers in top-tier international conferences and journals, including **IEEE TC**, **IEEE TCAD**, **ASPLOS**, **HPCA**, **MICRO**, **ICML**, and **ICLR**. I have served as a reviewer for leading conferences and journals, including HPCA, CVPR, ICCV, NeurIPS, AAAI, and IEEE TPDS.

Work Experience

- **The Hong Kong University of Science and Technology** HongKong
  Post-doctoral Fellow of Electrical and Computer Engineering *2026.7 -*

Education

- **Institute of Automation, Chinese Academy of Sciences** Beijing
  Pattern recognition and intelligent system, Ph.D Student *2021.9 - 2026.6*
  - **Overall GPA**: 3.92/4
- **Zhejiang University** Hangzhou
  Mechatronic Engineering, Bachelor *2017.9 - 2021.6*
  - **Overall GPA**: 4.66/5 (Ranking: 1/181)

Project Experience

- **An FPGA-Based LLM Edge Accelerator with Bitstream Quantization** Key Contributor
  2023.9-2024.7

  Responsible for the hardware implementation, including the design of KV Cache rearrangement units to support flexible KV Cache access, laying the foundation for bitstream-quantized model inference. Developed FP16 and INT8 MAC (Multiply-Accumulate) operators, ensuring alignment with GPU computation results and guaranteeing the correctness of bitstream-quantized model inference. Constructed DMA memory access pathways to efficiently utilize HBM bandwidth, addressing memory bottlenecks during LLM decoding. Additionally, handled overall accelerator simulation and verification, and implemented the system on the VHK158 board to validate the acceleration performance.

- **An FPGA-Based CNN Edge Accelerator with Model Quantization** Project Leader
  2021.2-2021.7

  Designed a CNN accelerator based on Ultra96 V2, achieving low-power, low-latency inference acceleration for edge-targeted object recognition tasks. The accelerator employs a systolic array architecture with custom-designed computation units, memory units, and dataflows for efficient data reuse. Additionally, a corresponding instruction set and compiler were developed to map CNN models onto the accelerator. On the algorithm side, an 8-bit fixed-point quantization of CNN was implemented to reduce storage and computation requirements, further accelerating the inference process.

Technical and Personal Skill

- **Programming Language:** Python, C/C++, CUDA.
- **Hardware Design Language:** Vivado HLS, Verilog.
- **Simulation / Emulation Tools:** Synopsys Design Compiler, Vivado, CACTI, DRAMSim.

Publication

1. **Zeyu Zhu**, Fanrong Li, Zitao Mo, Qinghao Hu, Gang Li, Zejian Liu, Xiaoyao Liang, Jian Cheng. A^2Q: Aggregation-Aware Quantization for Graph Neural Networks. In **International Conference on Learning Representations, (ICLR)**, 2023, CCF-A.

2. **Zeyu Zhu**, Fanrong Li, Gang Li, Zejian Liu, Zitao Mo, Qinghao Hu, Xiaoyao Liang, Jian Cheng. MEGA: A Memory-Efficient GNN Accelerator Exploiting Degree-Aware Mixed-Precision Quantization. In **High-Performance Computer Architecture, (HPCA)**, 2024, **CCF-A**.

3. **Zeyu Zhu**, Peisong Wang, Qinghao Hu, Gang Li, Xiaoyao Liang, Jian Cheng. FastGL: A GPU-Efficient Framework for Accelerating Sampling-Based GNN Training at Large Scale. In **ACM International Conference on Architectural Support for Programming Languages and Operating Systems, (ASPLOS)**, 2024, **CCF-A**.

4. **Zeyu Zhu**, Gang Li, Tielong Liu, Zitao Mo, Xiaoyao Liang, Jian Cheng. MATA: A Memory-Efficient Attention Accelerator for LLMs Exploiting Look-Back KV Cache Pruning. In **IEEE Transactions on Computers**, 2025, **CCF-A**.

5. **Zeyu Zhu**, Gang Li, Zitao Mo, Peisong Wang, Jian Cheng. KL-MoE: A Hierarchical MoE Pruning Framework Employing KL Divergence. In **Design Automation Conference, (DAC)**, 2026, **CCF-A**.

6. **Zeyu Zhu**, Gang Li, Zitao Mo, Peisong Wang, Jian Cheng. AdaOff: An Adaptive Offloading Framework for LLMs Inference on Local PCs. In **IEEE Transactions on Computer-Aided Design of Integrated Circuits & Systems (TCAD)**, 2026, **CCF-A**.

7. Wentao Hu*, **Zeyu Zhu***, Mingkuan Zhao, Zhenhua An, Yanbo Zhai, Shanhong Yu, Huilin Zhou, Xin Lai, Xiaoyan Zhu, Jiayin Wang. RaGEP: Rank-aware Geometric Expert Pruning for Mixture-of-Experts Language Models. In **International Conference on Machine Learning, (ICML)**, 2026, **CCF-A**.

8. Tielong Liu*, **Zeyu Zhu***, Gang Li, Zitao Mo, Chen BAI, Junwen Si, Peihuan Ni, Minnan Pei, Peisong Wang, Zhuoran Song, Xiaoyao Liang, Jian Cheng. FLUTE: A LUT-Based Floating-Point Engine for Efficient LLM Acceleration Exploiting Core-Tail Hierarchical Alignment. In **International Symposium on Microarchitecture, (MICRO)**, 2026, **CCF-A**.

9. Peihuan Ni*, **Zeyu Zhu***, Weifan Guan, Gang Li, Zitao Mo, Minnan Pei, Tielong Liu, Junwen Si, Qinghao Hu, Peisong Wang, Jian Cheng. ACE-VLA: Action Centric Evaluation for Token Pruning in Vision-Language-Action Models In **Conference on Robot Learning, (CoRL)**, 2026.

10. **Zeyu Zhu**, Gang Li, Zitao Mo, Peisong Wang, Jian Cheng. DALI: A Workload-Aware Offloading Framework for Efficient MoE Inference. **Under Review**.

11. Minnan Pei, Gang Li, Junwen Si, **Zeyu Zhu**, Zitao Mo, Peisong Wang, Zhuoran Song, Xiaoyao Liang, Jian Cheng. GCC: A 3DGS Inference Architecture with Gaussian-Wise and Cross-Stage Conditional Processing. In **MICRO**, 2025, **CCF-A**.

12. Ruiyang Chen, Zhuoran Song, Yicheng Zheng, **Zeyu Zhu**, Gang Li, Naifeng Jing, Xiaoyao Liang, Haibing Guan. HEAT: NPU-NDP HEterogeneous Architecture for Transformer-Empowered Graph Neural Networks. In **MICRO**, 2025, **CCF-A**.

13. Minnan Pei, Gang Li, **Zeyu Zhu**, Siting Wang, Junwen Si, Zhuoran Song, Yu Feng, Fangxin Liu, Xiaoyao Liang, Jian Cheng. DeGS: A Scalable 3DGS Architecture via Decoupled Workload Parsing and Reorganization. In **MICRO**, 2026, **CCF-A**.

14. Tielong Liu, Gang Li, Zitao Mo, **Zeyu Zhu**, Minnan Pei, Jian Cheng. Boosting the Performance of Tree-Based Speculative Decoding of LLMs on FPGAs. In **DATE**, 2026, **CCF-B**.

15. Peihuan Ni, Zitao Mo, Tielong Liu, Hongli Wen, **Zeyu Zhu**, Minnan Pei, Junwen Si, Weifan Guan, Peisong Wang, Qinghao Hu, Gang Li, Jian Cheng. APEX: Integer-only Non-linear Function Approximation for Efficient Cross-Modal Inference. In **DATE**, 2026, **CCF-B**.

Honor & Award

- **China University of Chinese Academy of Sciences Climbing Scholarship** *2024*
- **Merit Student of University of Chinese Academy of Sciences** *2023*
- **Outstanding Graduate of Zhejiang University** *2021*
- **Arawana Scholarship (Top 1%)** *2019*
- **Merit Student of Zhejiang University** *2018*

朱泽雨

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我是一名来自香港科技大学的博士后，导师为谢源教授。我博士毕业于中科院自动化研究所，导师为程健研究员。我的研究方向集中在关于LLM,GNN及CNN的**AI芯片设计、深度学习系统设计、 异构计算、网络模型压缩**等方面。目前，已经以第一作者/共一在国际顶级会议期刊上发表工作9篇，包括 **IEEE TC**, **IEEE TCAD**, **ASPLOS**, **HPCA**, **MICRO**, **ICML**, **ICLR**。并曾经担任 HPCA 的 PC， CVPR, ICLR, ICCV, NeurIPS, AAAI, TPDS 等会议和期刊审稿人。

工作经历

- **香港科技大学 (导师：谢源教授)** 香港
  电子及计算机工程系博士后研究员 *2026.7-*

教育经历

- **中科院自动化研究所** 北京
  模式识别与智能系统, 博士在读 *2021.9 - 2026.6*
  - **总GPA**: 3.92/4
- **浙江大学** 杭州
  机械电子工程，学士 *2017.9 - 2021.6*
  - **总GPA**: 4.66/5 (排名: **1/181**)

项目经历

- **基于FPGA及比特序列量化的LLM端侧加速器设计 (国家级) 2023.9-2024.7** 主要完成人

  负责硬件部分实现，设计了KV Cache的重排单元，支持灵活的KV Cache访存，为实现比特序列量化模型的推理奠定基础；实现了FP16与INT8乘加算子，实现了与GPU计算结果的对齐，保证了比特序列量化模型推理结果的正确性；构建 DMA访存通路，高效利用HBM带宽，缓解LLM解码过程中的访存墙问题；负责加速器整体仿真验证工作并制作系统，在 VHK158板子上验证加速效果。

- **基于FPGA及模型量化的CNN端侧加速器设计 2021.2-2021.7** 项目负责人

  设计了一款基于Ultra96 V2的CNN加速器，实现了基于CNN的目标识别任务的端侧低功耗，低延时推理加速。该加速器基于脉动阵列，设计定制的计算单元和存储单元及数据流，实现了数据的高效复用，并设计了配套的指令系统及编译器实现CNN到加速器的映射。此外，在算法侧，对CNN进行了定点8bit量化，以降低存储和计算需求，进一步加速推理过程。

实习经历

- **字节跳动Seed-3D团队** *2025.3-2025.5*

  参加3D贴图模型（DiT）加速，具体负责细粒度cuda Kernel的加速以及模型中attention部分加速。评测3D贴图模型生成效果。此外，负责解决在训练过程中使用Flash Attention加速技术由精度带来的训练不收敛问题。

专业技能

- **编程语言：** Python, C/C++, CUDA.
- **硬件设计语言：** Vivado HLS, Verilog.
- **仿真验证工具：** Synopsys Design Compiler, Vivado, CACTI, DRAMSim.

发表论文

1. **Zeyu Zhu**, Fanrong Li, Zitao Mo, Qinghao Hu, Gang Li, Zejian Liu, Xiaoyao Liang, Jian Cheng. A^2Q: Aggregation-Aware Quantization for Graph Neural Networks. In **International Conference on Learning Representations, (ICLR)**, 2023, CCF-A.

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11. Minnan Pei, Gang Li, Junwen Si, **Zeyu Zhu**, Zitao Mo, Peisong Wang, Zhuoran Song, Xiaoyao Liang, Jian Cheng. GCC: A 3DGS Inference Architecture with Gaussian-Wise and Cross-Stage Conditional Processing. In **MICRO**, 2025, **CCF-A**.

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荣誉及奖励

- **中国科学院大学攀登二等奖学金** *2024*
- **中国科学院大学三好学生** *2023*
- **浙江大学优秀毕业生** *2021*
- **金龙鱼奖学金 (Top 1%)** *2019*
- **浙江大学三好学生** *2018*